# The Sharp Smoothness Threshold for One-Dimensional Schatten-Class Commutators

Automated mathematical discovery run `fa_banach_001`
Lane 9, agent `agent_lane_09`, model GPT5.6

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**Status.** Candidate full result likely valid; human review needed. The construction includes the formerly delicate critical line.

## 1 Source question and answer

Wei and Zhang study Schatten membership of commutators $[T, M_b]$ for singular integral operators with minimally regular kernels [1]. In dimension one, for $1 < p < 2$, they ask on page 6 whether the sufficiency part of their Theorem 1.3 continues to hold for every $0 < \alpha < 1/2$: does

$$b \in \boldsymbol{B}_p(\mathbb{R}) \quad \Longrightarrow \quad [T, M_b] \in S_p(L_2(\mathbb{R}))$$

follow when the kernel has only the standard $\alpha$-Hölder regularity?

The answer has a sharp threshold. The proof in Section 5.1 of the source already gives sufficiency when $\alpha > 1/p - 1/2$. We construct counterexamples when $\alpha \le 1/p - 1/2$, including equality.

The standard kernel assumptions in dimension one are

$$|K(x,y)| \le \frac{C}{|x-y|} \tag{1}$$

and, whenever $|x-y| > 2|x-x'| > 0$,

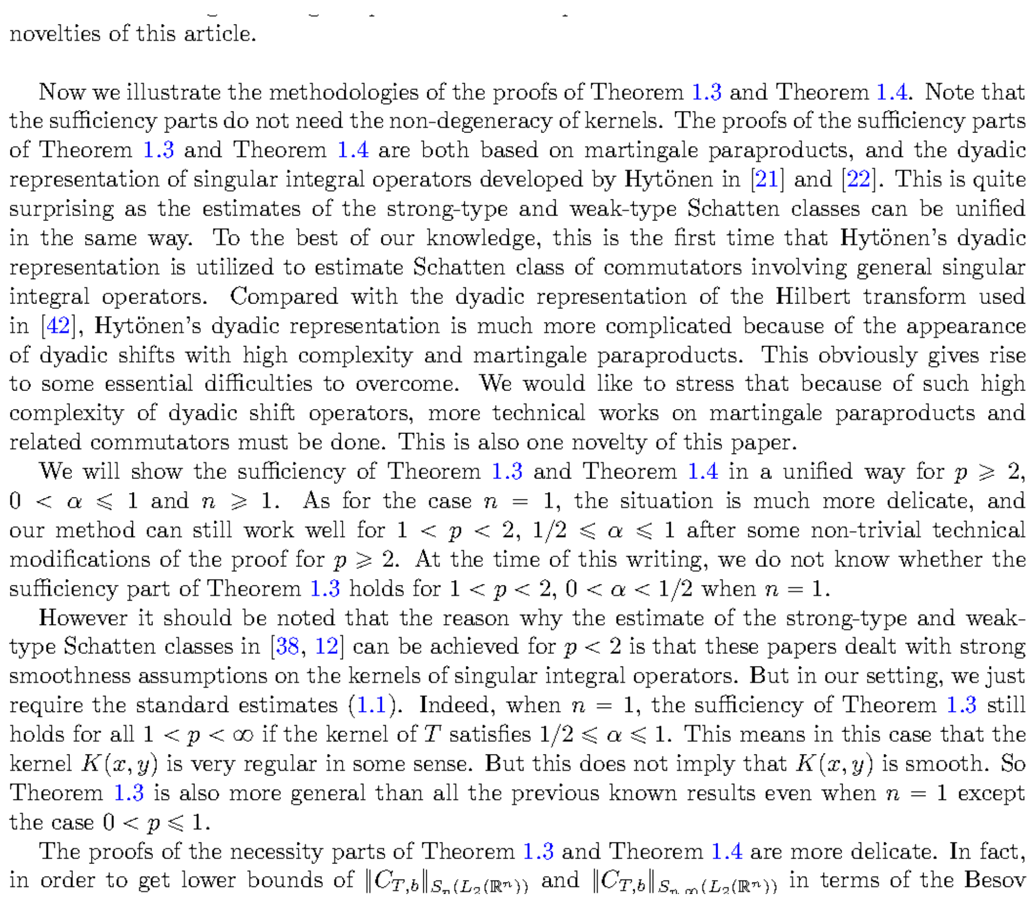
$$|K(x,y) - K(x',y)| + |K(y,x) - K(y,x')| \le C\frac{|x-x'|^\alpha}{|x-y|^{1+\alpha}}. \tag{2}$$

The source uses the seminorm

$$\|b\|^p_{\boldsymbol{B}_p(\mathbb{R})} = \int_{\mathbb{R}^2} \frac{|b(x)-b(y)|^p}{|x-y|^2}\, dx\, dy. \tag{3}$$

## 2 Why the endpoint is attainable

A single Hadamard block with $N$ inputs and $N$ outputs has $N$ equal singular values. At the critical line its Schatten mass is on exactly the same scale as the Besov cost of the symbol, so placing different scales in different spatial blocks cannot settle equality. The key is to let one fixed smooth symbol act on an orthogonal tower of wavelet scales.



novelties of this article.

Now we illustrate the methodologies of the proofs of Theorem 1.3 and Theorem 1.4. Note that the sufficiency parts do not need the non-degeneracy of kernels. The proofs of the sufficiency parts of Theorem 1.3 and Theorem 1.4 are both based on martingale paraproducts, and the dyadic representation of singular integral operators developed by Hytönen in [21] and [22]. This is quite surprising as the estimates of the strong-type and weak-type Schatten classes can be unified in the same way. To the best of our knowledge, this is the first time that Hytönen's dyadic representation is utilized to estimate Schatten class of commutators involving general singular integral operators. Compared with the dyadic representation of the Hilbert transform used in [42], Hytönen's dyadic representation is much more complicated because of the appearance of dyadic shifts with high complexity and martingale paraproducts. This obviously gives rise to some essential difficulties to overcome. We would like to stress that because of such high complexity of dyadic shift operators, more technical works on martingale paraproducts and related commutators must be done. This is also one novelty of this paper.

We will show the sufficiency of Theorem 1.3 and Theorem 1.4 in a unified way for $p \geqslant 2$, $0 < \alpha \leqslant 1$ and $n \geqslant 1$. As for the case $n = 1$, the situation is much more delicate, and our method can still work well for $1 < p < 2$, $1/2 \leqslant \alpha \leqslant 1$ after some non-trivial technical modifications of the proof for $p \geqslant 2$. At the time of this writing, we do not know whether the sufficiency part of Theorem 1.3 holds for $1 < p < 2$, $0 < \alpha < 1/2$ when $n = 1$.

However it should be noted that the reason why the estimate of the strong-type and weak-type Schatten classes in [38, 12] can be achieved for $p < 2$ is that these papers dealt with strong smoothness assumptions on the kernels of singular integral operators. But in our setting, we just require the standard estimates (1.1). Indeed, when $n = 1$, the sufficiency of Theorem 1.3 still holds for all $1 < p < \infty$ if the kernel of $T$ satisfies $1/2 \leqslant \alpha \leqslant 1$. This means in this case that the kernel $K(x, y)$ is very regular in some sense. But this does not imply that $K(x, y)$ is smooth. So Theorem 1.3 is also more general than all the previous known results even when $n = 1$ except the case $0 < p \leqslant 1$.

The proofs of the necessity parts of Theorem 1.3 and Theorem 1.4 are more delicate. In fact, in order to get lower bounds of $\|C_{T,b}\|_{S_p(L_2(\mathbb{R}^n))}$ and $\|C_{T,b}\|_{S_{p,\infty}(L_2(\mathbb{R}^n))}$ in terms of the Besov

Figure 1: The one-dimensional low-smoothness question in the source paper, page 6.

At scale $j$, couple $N_j \asymp 2^j$ compactly supported output wavelets to $N_j$ input wavelets by a Hadamard matrix with coefficient $\gamma_j = 2^{-j(1+\alpha)}$. The block then has $N_j$ singular values of size

$$\gamma_j \sqrt{N_j} \asymp 2^{-j(\alpha+1/2)}.$$

The kernel contribution has size $O(2^{-j\alpha})$ and Lipschitz constant $O(2^{j(1-\alpha)})$. Splitting the scale sum at $2^j|x - x'| \simeq 1$ gives a global $C^\alpha$ bound, with no logarithmic loss at equality.

## 3 Counterexample

**Theorem 1.** *Fix*

$$1 < p < 2, \qquad 0 < \alpha \leq \frac{1}{p} - \frac{1}{2}.$$

*There exist a bounded integral operator $T \in B(L_2(\mathbb{R}))$ with a kernel satisfying (1)–(2), and a real-valued bounded symbol $b \in \boldsymbol{B}_p(\mathbb{R})$, such that*

$$[T, M_b] \notin S_p(L_2(\mathbb{R})).$$

*The kernel is compactly supported, globally $C^\alpha$ in each variable, and supported a fixed positive distance from the diagonal. In fact $T$ is Hilbert–Schmidt.*

*Proof.* Choose a real compactly supported $C^1$ orthonormal wavelet $\omega$ on $\mathbb{R}$; compactly supported orthonormal wavelets of arbitrarily high regularity are standard. Write

$$\omega_{j,k}(x) = 2^{j/2}\omega(2^j x - k).$$

The family $\{\omega_{j,k} : j, k \in \mathbb{Z}\}$ is orthonormal. Fix disjoint open intervals

$$I = (0, 1), \qquad J = (4, 5).$$

For every sufficiently large $j$, there are $\asymp 2^j$ wavelets at scale $j$ whose supports are contained in $I$, and likewise in $J$. Consequently there are fixed integers $q, j_0 \geq 1$ such that, for every $j \geq j_0$, we may select index sets $E_j, F_j$ of cardinality

$$N_j = 2^{j-q}$$

with

$$\operatorname{supp}\omega_{j,k} \subset I \quad (k \in E_j), \qquad \operatorname{supp}\omega_{j,l} \subset J \quad (l \in F_j).$$

All selected output wavelets, across all scales, are orthonormal; the same is true of all selected input wavelets.

Let $H_{N_j} = (h^{(j)}_{kl})$ be a Sylvester Hadamard matrix, indexed by $E_j \times F_j$, so

$$H_{N_j} H^*_{N_j} = N_j I_{N_j}. \tag{4}$$

Put $\gamma_j = 2^{-j(1+\alpha)}$ and define

$$K(x, y) = \sum_{j=j_0}^{\infty} \gamma_j \sum_{k \in E_j} \sum_{l \in F_j} h^{(j)}_{kl} \omega_{j,k}(x)\omega_{j,l}(y). \tag{5}$$

We first verify the kernel estimates. Compact support of $\omega$ implies that, at a fixed point and scale, only a bounded number of wavelets can be nonzero. Since

$$\|\omega_{j,k}\|_\infty \leq C2^{j/2}, \qquad \|\omega'_{j,k}\|_\infty \leq C2^{3j/2},$$

the scale-$j$ summand $K_j$ in (5) satisfies

$$\|K_j\|_\infty \leq C2^{-j\alpha}, \qquad \sup_y \|\partial_x K_j(\cdot, y)\|_\infty + \sup_x \|\partial_y K_j(x, \cdot)\|_\infty \leq C2^{j(1-\alpha)}. \tag{6}$$

The first bound makes the kernel series uniformly convergent. If $\delta = |x - x'| > 0$, then (6) gives

$$|K_j(x, y) - K_j(x', y)| \leq C \min\{2^{-j\alpha},\, 2^{j(1-\alpha)}\delta\}.$$

Split the sum at the integer $m$ for which $2^m\delta \simeq 1$. The low scales form a geometric series bounded by

$$C\delta \sum_{j \leq m} 2^{j(1-\alpha)} \leq C\delta^\alpha,$$

and the high scales are bounded by

$$C \sum_{j > m} 2^{-j\alpha} \leq C\delta^\alpha.$$

The same argument applies in the second variable. Thus $K$ is globally $C^\alpha$ in each variable.

The kernel is supported in $I \times J$. Hence a nonzero value has $3 \le |x-y| \le 5$, and the uniform bound implies (1). For (2), use the global $C^\alpha$ estimate. If either value in a difference is nonzero and $|x-y| > 2|x-x'|$, the triangle inequality gives $|x-y| \le 10$. We may therefore insert the factor $10^{1+\alpha}/|x-y|^{1+\alpha}$. The transposed difference is identical.

Let $T$ be the integral operator with kernel $K$. Relative to the selected input and output wavelets, its scale-$j$ block is

$$A_j = \gamma_j H_{N_j}.$$

By (4), this block has precisely $N_j$ singular values equal to

$$s_j = \gamma_j \sqrt{N_j} = 2^{-q/2} 2^{-j(\alpha+1/2)}. \tag{7}$$

Different scale blocks have mutually orthogonal input and output spaces. Moreover,

$$\sum_{j \ge j_0} N_j s_j^2 = 2^{-2q} \sum_{j \ge j_0} 2^{-2j\alpha} < \infty.$$

Thus the kernel partial sums converge in Hilbert–Schmidt norm to $T$, and the uniformly convergent series (5) is indeed its integral kernel. In particular, $T$ is bounded.

Choose a real $b \in C_c^\infty((-1,2))$ with $b = 1$ on $I$. Then $b = 0$ on $J$. Smooth compact support and $p > 1$ imply directly from (3) that $b \in \boldsymbol{B}_p(\mathbb{R})$. Since every input wavelet is supported in $J$ and every output wavelet in $I$, we have

$$TM_b = 0, \qquad M_b T = T.$$

Therefore $[T, M_b] = -T$. Finally, (7) yields

$$\begin{aligned} \|[T, M_b]\|_{S_p}^p &= \sum_{j=j_0}^{\infty} N_j s_j^p \\ &= 2^{-q(1+p/2)} \sum_{j=j_0}^{\infty} 2^{j[1-p(\alpha+1/2)]} = \infty, \end{aligned}$$

because $1 - p(\alpha + 1/2) \ge 0$. This completes the counterexample, including the equality case. ☐

**Corollary 2** (Sharp threshold)**.** *Under the general kernel hypotheses of the source, for $1 < p < 2$ and $0 < \alpha < 1/2$, the implication*

$$b \in \boldsymbol{B}_p(\mathbb{R}) \Longrightarrow [T, M_b] \in S_p$$

*holds for every admissible $T$ exactly when $\alpha > 1/p - 1/2$.*

*Proof.* Sufficiency in the strict supercritical range is the estimate proved in Section 5.1 of [1]. Failure at and below the threshold is Theorem 1. ☐

# 4 Verification and literature check

The included exact script constructs Sylvester matrices and checks $H_N H_N^* = N I_N$ and the repeated block singular value formula. The scale summation and kernel estimates are exact geometric-series calculations.

Cheap run-index searches found no packet for arXiv:2411.05810. Bounded primary searches on July 22, 2026 used the arXiv id, title, authors, and the threshold phrase "$1/p - 1/2$"; they found the source and the authors' adjacent paper, but no later resolution of this question.

## 5 Human-review recommendation

Review as a likely valid full threshold result. The highest-value checks are: (i) selecting $\asymp 2^j$ compactly supported orthonormal wavelets inside each fixed interval; (ii) the low/high scale proof of the uniform $C^\alpha$ kernel estimate; (iii) the identification of the Hadamard block singular values across mutually orthogonal scales; and (iv) confirmation that the source's positive proof indeed applies for every $\alpha > 1/p - 1/2$ under exactly the displayed kernel hypotheses.

## References


[1] Z. Wei and H. Zhang, *Complex median method and Schatten class membership of commutators*, arXiv:2411.05810v2, 2024.